Pathwise Perturbation Particle Filtering with Anticipated-Shock Quadrature and Piecewise Localization

Ed (coauthor to be added)

February 18, 2026

Abstract

We formalize a likelihood-evaluation strategy for nonlinear DSGE models based on *pathwise* (a.k.a. extended-path) perfect-foresight solves and first-order linearizations around those paths. At each filtering step we augment the model with an auxiliary "anticipated-shock" object—a finite horizon of future structural innovations—that is used to approximate conditional expectations in the equilibrium conditions. Conditional on this auxiliary object (and on a discrete *localization index* that encodes the relevant piece of the state space or the binding pattern of constraints), the model is linear-Gaussian: the one-step transition is Gaussian with parameters given by the pathwise Jacobians, and the measurement equation is assumed linear with Gaussian noise. Integrating out the auxiliary object yields a Gaussian-mixture transition kernel. This structure enables a Rao–Blackwellized particle filter that marginalizes the high-dimensional continuous state with a Kalman filter and samples only the low-dimensional auxiliary object and discrete localization index.

We provide (i) a derivation of the conditional linear-Gaussian representation from the stacked equilibrium system via the implicit function theorem, (ii) sufficient conditions under which quadrature refinement converges to a well-defined mixture kernel and under which localization refinement yields a globally accurate piecewise-linear approximation, and (iii) stability-based bounds linking one-step kernel error to filtering and log-likelihood error. Finally, we propose a benchmark New Keynesian model with an effective lower bound (ELB) on the nominal rate and outline a comparison to OccBin and to the piecewise-linear continuous (PLC) solution and conditionally optimal particle filter (COPF) of Aruoba et al. (2021).

Keywords: nonlinear DSGE estimation; particle filter; Rao–Blackwellization; extended path; stochastic extended path; dynamic perturbation; sigma points; occasionally binding constraints; effective lower bound.
JEL: C11, C15, C61, E32.

1 Introduction

Likelihood-based estimation of nonlinear DSGE models is difficult because equilibrium dynamics are nonlinear, typically available only implicitly via a system of equilibrium conditions with conditional expectations, and frequently feature kinks from occasionally binding constraints (OBCs) such as an effective lower bound (ELB). A direct likelihood approach nests the resulting nonlinear state-space system inside a particle filter, and for parameter inference typically nests that filter inside an MCMC/SMC scheme (Kantas et al., 2015; Andrieu et al., 2010). Two practical bottlenecks dominate: computing a transition law that is accurate enough for likelihood-based inference, and controlling Monte Carlo variance of the likelihood estimator when the latent state is high-dimensional or the model has sharp nonlinearities.

A widely used strategy is to compute a global (or semi-global) approximation to the policy function (projection methods, value-function iteration, Smolyak/sparse grids, endogenous grid methods, discrete regime grids, etc.) and then filter the resulting nonlinear state-space system. This can be accurate, but it can be expensive in medium-to-large DSGEs and becomes particularly challenging near kinks where naive bootstrap particle filters can suffer severe weight degeneracy.

This paper formalizes an alternative approximation strategy geared specifically toward likelihood evaluation. Instead of approximating the policy function globally, we approximate the *local* dynamics in the neighborhood of *paths* that are relevant under the data. At each filtering step we (i) solve for a finite-horizon equilibrium path using a perfect-foresight/extended-path system conditional on a low-dimensional auxiliary object that represents a finite horizon of "anticipated" future shocks, and then (ii) compute a first-order expansion around this path. This is in the spirit of the Extended Path (EP) method of Fair and Taylor (1983) and its stochastic generalizations (Adjemian and Juillard, 2025), and is closely related to Dynamic Perturbation (Mennuni et al., 2025), which computes first-order expansions along simulated equilibrium paths.

The central novelty is to embed this pathwise linearization inside a *filter*. Conditional on the auxiliary "anticipated-shock" object (and, when needed, on a discrete localization index encoding a regime/binding pattern or a partition cell in state space), the one-step transition is approximately linear-Gaussian, with parameters obtained from Jacobians of the stacked equilibrium system. Integrating out the auxiliary object by sigma points or other quadrature yields a Gaussian-mixture (Gaussian-sum) transition kernel (Alspach and Sorenson, 1972). With linear-Gaussian measurement equations, this structure supports a Rao–Blackwellized particle filter (RBPF) that marginalizes a large continuous state block with a Kalman filter and samples only the low-dimensional auxiliary variables and discrete localization indices (Doucet et al., 2000).

The framework exposes a bias–variance–cost tradeoff. Finite-horizon solves, expectation approximations, and local linearization introduce approximation bias, while Rao–Blackwellization can substantially reduce Monte Carlo variance relative to a bootstrap particle filter (Gordon et al., 1993). Whether PPPF improves on global-solution-based likelihood evaluation depends on compute budgets, the effective dimension of the latent state, and how often the data drive the economy into regions where nonlinearities matter.

1.1 Contributions and organization

This paper contributes:

1. A derivation of a *conditional* linear-Gaussian approximation to DSGE transitions from stacked equilibrium conditions using the implicit function theorem, clarifying the link between EP/SEP, dynamic perturbation, and time-varying local linearizations along equilibrium paths.

2. A construction of a *Gaussian-mixture* transition kernel by integrating over an anticipated-shock auxiliary variable using quadrature (sigma points, Gauss–Hermite, sparse grids, Monte Carlo), and a discussion of when and how such quadrature can be used as a *probability mixture* inside an RBPF.

3. A detailed PPPF-mixture-RBPF recursion, including mixture handling (branching versus sampling), weight updates under general proposals, ESS-triggered resampling, and practical notes on localization indices for OBC regimes and piecewise "globalization" of local linearizations.

4. A benchmark New Keynesian model with an ELB and a reference implementation that reports runtime, ESS profiles, and likelihood-estimator variability, alongside impulse-response comparisons to OccBin and a discretized global stochastic benchmark.

The remainder of the paper is organized as follows. Section 2 introduces the nonlinear state-space model. Sections 3–5 derive the conditional linear-Gaussian approximation, its quadrature-induced mixture representation, and the role of discrete localization for globalization. Section 6 presents filtering and likelihood evaluation in detail. Sections 7–8 provide approximation-error decompositions and an explicit bias–variance–cost comparison to global-solution-based likelihood evaluation. Section 9 specifies and implements the New Keynesian ELB benchmark. Appendices collect derivations and quadrature details.

2 Related literature

This section situates PPPF relative to work on nonlinear DSGE likelihood evaluation, OBC solutions, and mixture/Rao–Blackwellized filtering.

2.1 Nonlinear DSGE likelihood evaluation and particle filtering

Likelihood-based estimation of DSGE models with nonlinear or non-Gaussian state dynamics typically relies on sequential Monte Carlo methods, often embedded inside parameter-estimation schemes (Kantas et al., 2015; Andrieu et al., 2010). In macroeconomic applications, particle-filter likelihoods have been used for estimation and model comparison in nonlinear DSGEs; see, e.g., Fernandez-Villaverde and Rubio-Ramirez (2007) and the textbook treatment in Herbst and Schorfheide (2016). Variance reduction and robustness near sharp nonlinearities have motivated methods such as tempered particle filtering (Herbst and Schorfheide, 2019) and auxiliary/proposal-based particle filters (Pitt and Shephard, 1999).

2.2 Occasionally binding constraints and piecewise approximations

OBC models introduce non-differentiabilities that complicate both solution and filtering. OccBin (Guerrieri and Iacoviello, 2015) and related finite-horizon regime-iteration methods produce piecewise-linear decision rules conditional on a binding pattern over a horizon, and have been widely used for ELB applications (Gust et al., 2017). Aruoba et al. (2021) develop piecewise-linear continuous (PLC) approximations for OBC DSGEs and a conditionally optimal particle filter (COPF) that exploits the resulting structure to reduce likelihood variance. PPPF is complementary: it uses pathwise linearizations and can incorporate an OBC binding pattern as a discrete localization index.

2.3 Extended-path, stochastic extended-path, and pathwise linearization

The extended-path method of Fair and Taylor (1983) solves a finite-horizon perfect-foresight system with a terminal condition, producing a time-varying local approximation to equilibrium dynamics that is often computationally attractive in medium-to-large models. Stochastic extended path (Adjemian and Juillard, 2025) improves on certainty equivalence by integrating over future shocks to better approximate conditional expectations. Dynamic perturbation (Mennuni et al., 2025) computes first-order expansions along simulated equilibrium paths and emphasizes accuracy along realized trajectories rather than at a fixed steady state. PPPF can be viewed as combining these ideas with a filtering recursion: build a pathwise linearization (conditional on a finite-dimensional shock-path proxy) and integrate it inside the likelihood evaluator.

2.4 Gaussian-sum filtering, sigma points, and Rao–Blackwellization

Gaussian-sum (mixture) representations of nonlinear filtering problems date back to Alspach and Sorenson (1972). Rao–Blackwellized particle filters (Doucet et al., 2000) exploit conditional linear-Gaussian structure to integrate out continuous components analytically via Kalman filtering, reducing Monte Carlo variance relative to a full bootstrap filter (Gordon et al., 1993). Sigma-point quadrature and related deterministic integration rules are widely used to approximate Gaussian expectations and underlie unscented filtering (Julier and Uhlmann, 2004). PPPF uses such quadrature not as a single-Gaussian moment-matching device, but to produce an explicit mixture kernel that can be handled either by branching or by sampling mixture indices within an RBPF.

3 Nonlinear state-space model

3.1 Latent state and measurement

Let $(x_t)_{t\geq 0}$ be an $\mathbb{R}^{n_x}$-valued latent state and $(y_t)_{t\geq 1}$ observed data. We assume a linear measurement equation with Gaussian noise:

$$y_t = Hx_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R), \tag{1}$$

where H and R may depend on parameters θ.

3.2 Equilibrium transition and Gaussian shocks

The state transition is induced by DSGE equilibrium conditions and structural innovations. For exposition we write it as a Markov mapping

$$x_{t+1} = F(x_t, \varepsilon_{t+1}; \theta), \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \Sigma), \tag{2}$$

even though in DSGEs F is usually not available in closed form.

Let $K_\theta(x, \cdot)$ denote the true Markov kernel of $x_{t+1} \mid x_t = x$ induced by (2).

4 From stacked equilibrium conditions to a conditional linear-Gaussian law

This section derives the object the filter will use: a Gaussian approximation to $K_\theta(x_t, \cdot)$ conditional on an auxiliary object (anticipated shocks) and a discrete localization index.

4.1 A finite-horizon "stochastic perfect foresight" system

Fix a horizon $H \geq 2$. Let

$$\omega_t = (\varepsilon^a_{t+1}, \varepsilon^a_{t+2}, \ldots, \varepsilon^a_{t+H-1}) \in \mathbb{R}^d, \qquad d = (H-1)n_\varepsilon,$$

collect a sequence of *anticipated* shocks over the horizon. We treat ω_t as an auxiliary variable drawn from the Gaussian law induced by the shock process:

$$\omega_t \sim \mathcal{N}(0, \Omega), \qquad \Omega = \mathrm{diag}(\Sigma, \ldots, \Sigma).$$

Conceptually, ω_t is a numerical device used to approximate conditional expectations, not an additional structural shock.

Let $X_{t:t+H-1} = (x_t, x_{t+1}, \dots, x_{t+H-1})$ and define a stacked residual mapping

$$\mathcal{R}(X_{t:t+H-1}; x_t, \omega_t, \theta) = 0$$

encoding equilibrium conditions from t through $t+H-2$ plus a terminal condition at $t+H-1$ (e.g. $x_{t+H-1} \approx x^\star$). In practice this is the system solved in EP/SEP and in perfect-foresight solvers (Fair and Taylor, 1983; Adjemian and Juillard, 2025). We denote by

$$\bar{X}_{t:t+H-1} = \bar{X}(x_t, \omega_t; \theta)$$

a solution of the stacked system.

4.2 Local perturbations around the path

Define deviations $\tilde{x}_s = x_s - \bar{x}_s$ for $s = t, \dots, t+H-1$. Linearizing $\mathcal{R}$ at $\bar{X}$ yields a sparse block system linking $\tilde{x}_{t:t+H-1}$ to *unexpected* innovations. To keep the derivation transparent, we focus on the one-step mapping from (x_t, ε_{t+1}) to x_{t+1}. Let ε_{t+1} be the *unexpected* shock realization at $t+1$ (mean zero). Then the linearized system implies

$$\tilde{x}_{t+1} = A_t(x_t, \omega_t)\, \tilde{x}_t + B_t(x_t, \omega_t)\, \varepsilon_{t+1}, \tag{3}$$

where A_t, B_t are obtained from the Jacobian of the stacked system (details in Appendix A).

Equation (3) yields a *conditional* linear-Gaussian transition:

$$x_{t+1} \mid (x_t, \omega_t) \;\approx\; \mathcal{N}(m(x_t, \omega_t),\, Q(x_t, \omega_t)), \tag{4}$$

with

$$m(x_t, \omega_t) = \bar{x}_{t+1} + A_t(x_t, \omega_t)\,(x_t - \bar{x}_t), \qquad Q(x_t, \omega_t) = B_t(x_t, \omega_t)\, \Sigma\, B_t(x_t, \omega_t)^\top.$$

Remark 1 (Relation to dynamic perturbation and SEP). Dynamic perturbation (Mennuni et al., 2025) also computes first-order expansions along equilibrium paths. SEP (Adjemian and Juillard, 2025) integrates over future shocks to better approximate conditional expectations than certainty equivalence. The PPPF construction can be viewed as: (i) using a SEP-style draw/quadrature node ω_t to build the reference path, (ii) computing a dynamic-perturbation-style linearization around that path, and (iii) integrating out ω_t inside a filter.

5 Integrating out anticipated shocks

5.1 Gaussian-mixture transition kernel

Integrating out ω_t yields an approximate transition kernel

$$\widehat{K}_\theta(x_t,\, \mathrm{d}x_{t+1}) = \int \mathcal{N}(\,\mathrm{d}x_{t+1};\, m(x_t, \omega), Q(x_t, \omega))\; \phi(\omega; 0, \Omega)\; \mathrm{d}\omega. \tag{5}$$

This is a Gaussian mixture (Gaussian-sum) transition (Alspach and Sorenson, 1972). In practice we replace the integral by a K-node rule (sigma points, Gauss–Hermite, sparse grids, quasi-Monte Carlo, or Monte Carlo):

$$\widehat{K}_{\theta,K}(x_t,\, \mathrm{d}x_{t+1}) = \sum_{k=1}^{K} w_k\, \mathcal{N}\Big(\,\mathrm{d}x_{t+1};\, m(x_t, \omega^{(k)}), Q(x_t, \omega^{(k)})\Big), \qquad \sum_{k=1}^{K} w_k = 1. \tag{6}$$

5.2 Sigma points as deterministic "anticipated shock" draws

For $\omega \sim \mathcal{N}(0, \Omega) \in \mathbb{R}^d$, the basic unscented transform uses $2d + 1$ sigma points and associated weights that match mean and covariance and are exact for polynomials up to degree three (for symmetric choices) (Julier and Uhlmann, 2004). Sigma points are attractive when d is small (e.g. a small number of shocks and short horizon).

6 Discrete localization and "globalization" of the approximation

Sigma-point refinement controls the numerical integration error in (5), but does *not* by itself eliminate the local-linearization and horizon/terminal-condition errors. This section introduces a second approximation axis that can yield a genuine global refinement scheme.

6.1 Localization index: partition cells or regime/binding patterns

Let $s_t \in \{1, \dots, J\}$ be a discrete localization index. Two cases are especially relevant:

1. **State-space partition.** Partition a compact set $\mathcal{X} \subset \mathbb{R}^{n_x}$ into cells $\{\mathcal{X}_j\}_{j=1}^J$ and choose anchors $x^{(j)} \in \mathcal{X}_j$. For each anchor (and each quadrature node $\omega^{(k)}$) compute and store $(\bar{x}_{t+1}^{(j,k)}, A^{(j,k)}, B^{(j,k)})$. At runtime, map each x_t to its cell index $s_t = j$ and use the corresponding precomputed linearization parameters. This turns online path solving into offline precomputation plus fast lookup/interpolation.
2. **Occasionally binding constraints (OBCs).** Let s_t encode the binding/slack pattern of a constraint over the horizon (a regime path). OccBin (Guerrieri and Iacoviello, 2015) selects a regime path by iterating on guesses until the constraint is satisfied. In a filter, the regime can be treated as latent: particles can branch over regimes and the filter integrates over them. Aruoba et al. (Aruoba et al., 2021) construct globally piecewise-linear and continuous decision rules and derive a conditionally optimal particle filter (COPF) that exploits this structure. Our framework nests the idea of treating the binding pattern as a discrete index that makes the conditional system linear.

6.2 A simple approximation theorem for localization refinement

To formalize "convergence to global" under partition refinement, we state a generic result. For simplicity, consider a setting where $(m(x, \omega), Q(x, \omega))$ is smooth in x for each ω (away from kinks). Let $\widehat{m}_J(x, \omega)$ and $\widehat{Q}_J(x, \omega)$ be piecewise-linear approximations built from anchors with mesh size

$$\Delta_J = \max_{j \le J} \sup_{x \in \mathcal{X}_j} \|x - x^{(j)}\|.$$

Assumption 1 (Smoothness in x)**.** *For each ω in the support of the quadrature rule and all $x \in \mathcal{X}$, the maps $x \mapsto m(x, \omega)$ and $x \mapsto Q(x, \omega)$ are twice continuously differentiable with uniformly bounded second derivatives.*

Proposition 1 (Localization refinement implies uniform parameter convergence)**.** *Under the smoothness assumption, there exists $C < \infty$ such that*

$$\sup_{x \in \mathcal{X}} \|m(x, \omega) - \widehat{m}_J(x, \omega)\| \le C\Delta_J^2, \qquad \sup_{x \in \mathcal{X}} \|Q(x, \omega) - \widehat{Q}_J(x, \omega)\| \le C\Delta_J^2,$$

for each fixed ω, provided $\widehat{m}_J, \widehat{Q}_J$ are constructed by first-order Taylor (or equivalent local linear) approximations on each cell. Hence if $\Delta_J \to 0$ as $J \to \infty$, then the approximation error vanishes uniformly.

Remark 2. This is the standard "mesh $\to 0$" globalization logic: piecewise linearization converges for smooth maps. For OBCs, the map is typically only piecewise smooth and the rate can degrade near the kink. In that case one refines the partition more aggressively near the kink set, or treats the kink via an explicit discrete regime as in OccBin and PLC/COPF.

7 Filtering and likelihood evaluation

7.1 Gaussian-sum representation

Under (6), the approximate transition can be written by introducing a discrete index $k_t \in \{1, \dots, K\}$ with $\Pr(k_t = k) = w_k$:

$$x_{t+1} \mid (x_t, k_t = k) \sim \mathcal{N}(m_k(x_t), Q_k(x_t)).$$

With localization s_t, the index becomes (s_t, k_t).

7.2 Rao–Blackwellized particle filter

Because the measurement is linear-Gaussian and the transition is conditionally Gaussian given (s_t, k_t), we can marginalize the continuous state by Kalman filtering conditional on each particle's discrete history. A Rao–Blackwellized particle filter (RBPF) samples (s_t, k_t) (or, in a continuous version, samples ω_t) and updates the conditional Gaussian state distribution analytically (Doucet et al., 2000).

7.3 Algorithmic detail: PPPF-mixture-RBPF recursion

This subsection records the filtering recursion explicitly. For ease of exposition we assume the mixture components are locally affine in x_t:

$$x_{t+1} \mid (x_t, s_t = s, k_t = k) \;\approx\; A_t^{(s,k)} x_t + a_t^{(s,k)} + \xi_{t+1}^{(s,k)}, \qquad \xi_{t+1}^{(s,k)} \sim \mathcal{N}(0, Q_t^{(s,k)}),$$

and the measurement equation is linear-Gaussian as in (1). Each RBPF particle stores a Kalman-filter state (mean/covariance) for the continuous state conditional on its discrete history.

A practitioner-oriented algorithm. Given N particles and data $y_{1:T}$, an RBPF likelihood evaluator can be summarized as:

1. Initialize each particle i with a Gaussian prior $(\mu_0^{(i)}, P_0^{(i)})$ and weight $w_0^{(i)} = 1/N$.
2. For each $t = 1, \dots, T$:
    (a) Build (or update) a mixture approximation to the one-step transition, producing weights $w_t^{(s,k)}$ and component parameters $(A_t^{(s,k)}, a_t^{(s,k)}, Q_t^{(s,k)})$. In practice this can be done using a reference state (e.g. a weighted mean of particles) to avoid solving an extended-path system per particle.
    (b) For each particle i:

i. Sample a discrete index $j_t^{(i)} = (s_t^{(i)}, k_t^{(i)})$ from a proposal $q_t(\cdot)$ (the simplest choice is the prior $q_t(j) = w_t^{(j)}$).

ii. Kalman predict under the sampled component: $(\mu_{t|t-1}^{(i)}, P_{t|t-1}^{(i)})$.

iii. Kalman update given y_t, obtaining $(\mu_t^{(i)}, P_t^{(i)})$ and the predictive density $p(y_t \mid y_{1:t-1}, j_t^{(i)}, i)$.

iv. Update the particle weight by multiplying by the predictive density and the proposal correction factor:

$$w_t^{(i)} \propto w_{t-1}^{(i)}\, p(y_t \mid y_{1:t-1}, j_t^{(i)}, i)\, \frac{w_t^{(j_t^{(i)})}}{q_t(j_t^{(i)})}.$$

(c) Normalize weights, compute ESS, and resample if ESS falls below a threshold (e.g. $0.5N$).

(d) Accumulate the log-likelihood estimate via the (stable) log-sum-exp of unnormalized weights.

The key advantage of RBPF is that the Kalman step analytically integrates out the continuous state conditional on the discrete history, often providing substantial variance reduction. More sophisticated proposals (auxiliary particle filters) can improve ESS further by conditioning q_t on y_t (Pitt and Shephard, 1999).

7.4 Algorithmic detail: mixture handling and quadrature choices

A key practical point is *mixture handling*. Given a mixture transition $\sum_k w_k \mathcal{N}(m_k, Q_k)$, averaging the means m_k and covariances Q_k to obtain a single Gaussian is generally incorrect; it destroys tail behavior and Jensen effects. Instead, one must treat the mixture either by branching (expanding particles) or by sampling k and accounting for the mixture density in the proposal.

In the RBPF setting, branching corresponds to spawning children for each (s, k) and then resampling. Sampling corresponds to drawing (s, k) from their prior weights (or an auxiliary proposal) and updating particle weights accordingly.

Nonnegative weights for probability mixtures. Sigma-point rules (e.g. unscented transform) are often presented as deterministic quadrature and may assign negative weights in some parameterizations. Negative weights can be acceptable for moment computation, but they are incompatible with *sampling mixture indices* inside an RBPF. When PPPF is implemented as a probability mixture, the quadrature rule must yield nonnegative weights (e.g. a positive-weight sigma-point/cubature rule, Gauss–Hermite with positive weights, sparse grids with positive weights on tensorized rules, or Monte Carlo with equal weights). Alternatively, one can implement the integral over ω_t by branching and aggregating weights without interpreting quadrature weights as sampling probabilities.

Localization indices for kinks and regimes. For OBC models, a natural localization index is the binding pattern over the EP horizon (as in OccBin), which yields piecewise-linear dynamics conditional on the regime. More generally, localization can represent a partition of the state space, refined adaptively near kinks, so that first-order Taylor expansions are accurate within each cell. Section 5 formalizes this "globalization" logic.

8 Approximation theory: kernel, filter, and likelihood

8.1 Quadrature refinement converges to the mixture kernel

Let $\widehat{K}_\theta$ be the mixture kernel (5) and $\widehat{K}_{\theta,K}$ its K-node quadrature approximation (6).

Assumption 2 (Quadrature consistency)**.** *The quadrature rules* $(\omega^{(k)}, w_k)_{k=1}^K$ *are consistent for the Gaussian measure* $\mathcal{N}(0, \Omega)$*:*

$$\sum_{k=1}^{K} w_k h(\omega^{(k)}) \to \int h(\omega)\, \phi(\omega; 0, \Omega)\, \mathrm{d}\omega \quad \textit{for all bounded continuous } h.$$

Proposition 2 (Weak convergence of $\widehat{K}_{\theta,K} \to \widehat{K}_\theta$)**.** *Assume* $(m(x, \omega), Q(x, \omega))$ *is continuous in* ω *for each fixed* x *and that dominated convergence applies to the Gaussian density integrand. Then for each fixed* x *and each bounded continuous test function* ψ*,*

$$\int \psi(x')\, \widehat{K}_{\theta,K}(x, \mathrm{d}x') \to \int \psi(x')\, \widehat{K}_\theta(x, \mathrm{d}x').$$

Remark 3. This convergence holds to the mixture kernel $\widehat{K}_\theta$, not to the true DSGE kernel K_θ. Closing the remaining gap requires controlling the local-linearization error, horizon truncation/terminal-condition error, and expectation-approximation error in the underlying stochastic perfect-foresight step.

8.2 Kernel error decomposition

Write the total kernel error as

$$\|K_\theta(x, \cdot) - \widehat{K}_{\theta,K,J}(x, \cdot)\| \le \underbrace{\|K_\theta - \widetilde{K}_{\theta,H,p}\|}_{\text{horizon/expectation approximation}} + \underbrace{\|\widetilde{K}_{\theta,H,p} - \widehat{K}_\theta\|}_{\text{local linearization}} + \underbrace{\|\widehat{K}_\theta - \widehat{K}_{\theta,K}\|}_{\text{quadrature}} + \underbrace{\|\widehat{K}_{\theta,K} - \widehat{K}_{\theta,K,J}\|}_{\text{localization}},$$

where $\widetilde{K}_{\theta,H,p}$ denotes the kernel induced by the *exact* finite-horizon stochastic perfect-foresight solution with p periods of expectation integration (no linearization), and $\widehat{K}_{\theta,K,J}$ denotes the practical PPPF kernel with K quadrature nodes and localization mesh index J.

8.3 From kernel error to filter and likelihood error

Let $\delta = \sup_{x \in \mathcal{X}} \|K_\theta(x, \cdot) - \widehat{K}_{\theta,K,J}(x, \cdot)\|_{\mathrm{TV}}$ be the worst-case one-step kernel error. Under standard mixing/forgetting conditions for hidden Markov models, errors in the transition kernel propagate to errors in filtering distributions and predictive likelihoods; see, e.g., Del Moral (2004); Whiteley (2013); Kantas et al. (2015).

Assumption 3 (Uniform mixing and bounded likelihood)**.** *There exists* $\epsilon \in (0, 1)$ *and a probability measure* ν *on* $\mathcal{X}$ *such that for all* $x \in \mathcal{X}$ *and measurable* A*,*

$$K_\theta(x, A) \ge \epsilon \nu(A),$$

and the measurement density satisfies $0 < g_{\min} \le g_\theta(y \mid x) \le g_{\max} < \infty$ *for all* $x \in \mathcal{X}$ *and realized* y_t*.*

Theorem 1 (Filter perturbation bound (standard form))**.** *Under the mixing and bounded-likelihood assumption, there exist constants* $C < \infty$ *and* $\rho \in (0, 1)$ *such that the filtering distributions satisfy*

$$\|\pi_{\theta,t} - \widehat{\pi}_{\theta,t}\|_{\mathrm{TV}} \le C \sum_{s=1}^{t} \rho^{t-s}\, \delta, \qquad t \ge 1.$$

Corollary 1 (Log-likelihood perturbation bound (sketch))**.** *Under the same conditions, there exists* $C_\ell < \infty$ *such that*

$$|\log p_\theta(y_{1:T}) - \log \widehat{p}_\theta(y_{1:T})| \le C_\ell\, T\, \delta$$

up to geometric forgetting factors.

9 When can PPPF be "better" than a global solution?

Let ℓ_θ denote the *true* log-likelihood and let $\widehat{\ell}_\theta^{(A)}$ be a particle-based estimator using approximation strategy $A \in \{\text{glob}, \text{pppf}\}$. Under a compute budget C, the feasible particle count is roughly $N_A \approx C/c_A$, where c_A is cost per particle per period.

A standard decomposition is

$$\mathbb{E}\left[(\widehat{\ell}_\theta^{(A)} - \ell_\theta)^2\right] = \underbrace{\left(\text{Bias}_A\right)^2}_{\text{approximation bias}} + \underbrace{\frac{\sigma_A^2}{N_A}}_{\text{Monte Carlo variance}}.$$

Rao–Blackwellization reduces σ^2_{pppf} by integrating out (part of) the latent state analytically (Doucet et al., 2000). A sufficient "dominance" condition is therefore:

$$(\text{Bias}_{\text{pppf}})^2 + \sigma^2_{\text{pppf}}\frac{c_{\text{pppf}}}{C} < (\text{Bias}_{\text{glob}})^2 + \sigma^2_{\text{glob}}\frac{c_{\text{glob}}}{C}.$$

This inequality makes explicit why PPPF can lose in small models (high c_{pppf}, weak RB gains) but win in large models where the global solution is expensive and RB gains are large.

10 Benchmark: a New Keynesian model with an ELB

This section specifies a concrete benchmark that directly connects to the OccBin and PLC/COPF literatures.

10.1 Core equilibrium conditions

We consider a canonical three-equation New Keynesian model with an effective lower bound (ELB) on the nominal rate. Let x_t denote the output gap, π_t inflation, and i_t the nominal interest rate (deviations from steady state, appropriately scaled). A standard log-linearized NK system is

$$x_t = \mathbb{E}_t x_{t+1} - \sigma^{-1}\Big(i_t - \mathbb{E}_t\pi_{t+1} - r_t^n\Big), \tag{7}$$

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa x_t, \tag{8}$$

$$i_t^\star = \phi_\pi\pi_t + \phi_x x_t + \nu_t, \tag{9}$$

$$i_t = \max\{\underline{i},\ i_t^\star\}. \tag{10}$$

Here r_t^n is a natural-rate (or preference) shock and ν_t a monetary policy shock; both follow AR(1) laws with Gaussian innovations. Appendix C provides a brief microfoundation and discusses common variations (habit, indexation, wage rigidities).

The occasionally binding ELB constraint (10) introduces a kink and makes the decision rules nonlinear and piecewise smooth.

10.2 State-space representation and measurement

A minimal latent state can be

$$x_t^{\text{state}} = (x_t, \pi_t, i_t, r_t^n, \nu_t)^\top,$$

with observables

$$y_t = (x_t^{\text{obs}}, \pi_t^{\text{obs}}, i_t^{\text{obs}})^\top, \qquad y_t = Hx_t^{\text{state}} + \eta_t.$$

Because the ELB affects the transition, likelihood evaluation requires a nonlinear filter unless the solution imposes a structure that can be exploited.

10.3 Competing methods

We propose comparing three likelihood-evaluation strategies:

1. **OccBin (piecewise linear perturbation) + bootstrap PF / inversion filter.** OccBin (Guerrieri and Iacoviello, 2015) computes a piecewise linear solution by iterating on the binding pattern of (10) over a finite horizon. Likelihood evaluation can be done by a bootstrap particle filter, and several papers use inversion-filter variants when the observables identify shocks (or approximately identify them).
2. **Piecewise-linear continuous (PLC) decision rules + COPF.** Aruoba et al. (2021) construct globally piecewise-linear and continuous decision rules for OBC models and derive a *conditionally optimal* particle filter (COPF) that exploits the piecewise-linear structure to reduce likelihood variance. In our reference implementation we include a PLC-style baseline by interpolating a discretized global benchmark decision rule and using a conditionally informed (local-linearization) proposal for the exogenous states; this is intended as a computationally lightweight proxy for the PLC/COPF logic.
3. **PPPF with anticipated-shock quadrature and regime localization.** In the PPPF approach, we treat the ELB binding pattern over the EP horizon as part of the localization index s_t and integrate over short-horizon anticipated shocks ω_t by sigma points/quadrature. Conditional on (s_t, ω_t) the system is linear, enabling Rao–Blackwellization.

10.4 Proposed evaluation metrics

Because particle filters yield stochastic likelihood approximations, the key objects for comparison are:

- runtime per likelihood evaluation,
- Monte Carlo variance of the log-likelihood estimate (across repeated PF runs on fixed data),
- effective sample size (ESS) profiles and frequency of resampling,
- impulse responses: nonlinear "true" (high-accuracy) vs OccBin/PLC vs PPPF.

This benchmark is designed to highlight cases where PPPF can plausibly outperform a naive bootstrap PF: near the ELB kink, the conditional Gaussian structure and regime-aware localization can provide substantially better proposals and variance reduction.

10.5 Reference implementation and experimental design

We provide a minimal, reproducible reference implementation in the companion code directory `pppf-mixture-rbpf/`. The implementation includes:

1. an analytic Jensen-style validation (Burnside recursion) comparing certainty-equivalence versus sigma-point integration against a closed-form benchmark, and
2. a New Keynesian ELB benchmark comparing (i) an OccBin-style piecewise solution paired with a bootstrap particle filter and (ii) a PPPF-style Gaussian-mixture transition paired with an RBPF.

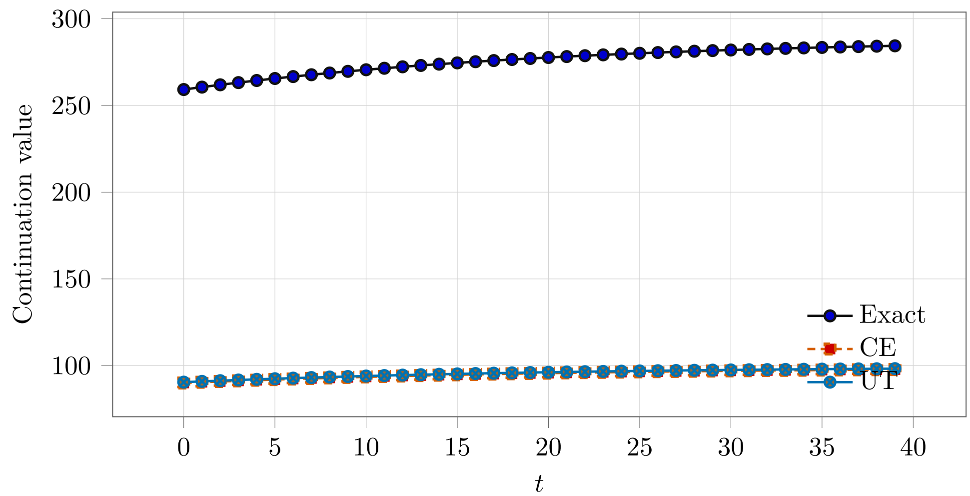

Figure 1: Burnside recursion validation (reference implementation): exact benchmark versus certainty equivalence (CE) and unscented-transform (UT) quadrature.

In the NK–ELB experiment, the OccBin extended-path horizon is $H = 20$. We solve the discretized global benchmark (Tauchen grid) and simulate a dataset of length $T = 250$ from that benchmark, adding small independent measurement noise on (π_t, x_t, i_t). For likelihood evaluation we use $N = 256$ particles and repeat each filter run $R = 30$ times with different random seeds to measure Monte Carlo variability. All artifacts (CSV + plots) are written under `pppf-mixture-rbpf/output/`.

Exact likelihood for the discretized global benchmark (HMM forward algorithm). The discretized global benchmark used in this experiment induces a finite-state hidden Markov model (HMM) for the exogenous state. Let $s_t \in \{1, \dots, S\}$ index the Tauchen grid point for (r_t^n, ν_t) and let $P \in \mathbb{R}^{S\times S}$ be the resulting transition matrix. Solving the discretized global model delivers an expectation-consistent mapping from each grid point to observables $g(s) = (\pi(s), x(s), i(s))^\top$. With Gaussian measurement noise, $y_t \mid s_t \sim \mathcal{N}(g(s_t), R)$, the log-likelihood of the discretized benchmark can be computed exactly by the standard forward recursion:

$$\alpha_{t|t-1}(s') = \sum_{s=1}^{S} \alpha_{t-1}(s)\, P(s, s'), \qquad \tilde{\alpha}_t(s) = \alpha_{t|t-1}(s)\, p(y_t \mid s), \qquad z_t = \sum_{s=1}^{S} \tilde{\alpha}_t(s),$$

with $\alpha_t(s) = \tilde{\alpha}_t(s)/z_t$ and $\log p(y_{1:T}) = \sum_{t=1}^{T} \log z_t$. This likelihood is exact for the *discretized benchmark* (grid + global policy functions), not for the underlying continuous-state DSGE.

10.6 Illustrative results

We begin with an analytic Jensen-style validation based on Burnside's recursion, where an exact benchmark is available. Figure 1 reports impulse responses for the continuation value under the exact benchmark, a certainty-equivalence approximation, and a sigma-point (UT) one-step quadrature approximation. The UT approximation tracks curvature effects better than certainty equivalence, illustrating why explicit integration over a low-dimensional shock object can matter even when the underlying state is Gaussian.

Method	Mean loglik	SD loglik	Mean runtime (ms)	SD runtime (ms)	Mean E
Global discrete benchmark (exact)	2495.0	0.0	7.2	0.0	
OccBin + bootstrap PF	2199.4	8.9	102.0	1.3	2
PPPF-mixture + RBPF	2130.0	0.5	185.7	3.6	18
PLC (interp) + COPF-style PF	2415.3	3.4	40.7	1.9	13

Table 1: NK–ELB benchmark (reference implementation): likelihood variability across $R = 30$ repeated particle-filter runs on fixed data, with $T = 250$ and $N = 256$ particles. The "global discrete benchmark" is exact for the discretized model.

As an additional sanity check in a setting where all methods should agree, we validate the filtering layer on a one-dimensional linear-Gaussian state-space model with known Kalman-filter likelihood. In that case, an RBPF with a single mixture component ($K = 1$) coincides with the Kalman filter and matches the exact log-likelihood to machine precision. A bootstrap particle filter, which replaces analytic Gaussian integration with Monte Carlo integration, approaches the same likelihood as the particle count increases.

We then turn to the NK–ELB benchmark. Table 1 reports summary statistics from the reference implementation. We simulate data from the discretized global benchmark and compute its exact likelihood by the HMM forward algorithm described above. Relative to OccBin + bootstrap PF, the PPPF-mixture-RBPF delivers much higher ESS and substantially lower Monte Carlo variability of the likelihood estimator, consistent with the variance-reduction role of Rao–Blackwellization. We also report a PLC-style baseline obtained by interpolating the global (discretized) benchmark policies on the (r^n, ν) grid and filtering only the exogenous states with a conditionally informed proposal based on local linearization; this provides a practical point of contact with the PLC/COPF perspective of Aruoba et al. (2021) while keeping the implementation minimal. The tradeoffs are transparent: Rao–Blackwellization and observation-informed proposals can dramatically reduce Monte Carlo variance (high ESS, low SD across repeated runs), but approximation bias remains whenever the transition kernel used by the likelihood evaluator deviates from the data-generating benchmark (finite-horizon truncation, local linearization, low-dimensional anticipated-shock objects, or grid discretization).

Figure 2 reports impulse responses near the ELB kink, comparing OccBin and PPPF to a discretized global stochastic benchmark (expectation-consistent equilibrium on a Tauchen grid). In this configuration, OccBin tracks the global benchmark more closely than PPPF in IRF RMSE, while PPPF exhibits substantially higher ESS (hence lower Monte Carlo variability) in likelihood evaluation. The lower PPPF likelihood relative to OccBin in Table 1 reflects approximation bias rather than particle-filter variance: PPPF is evaluating a different approximate kernel, and in the current prototype the anticipated-shock proxy and local linearization are not yet rich enough to match the discretized global benchmark in this calibration.

11 Conclusion

PPPF replaces a global nonlinear approximation of DSGE policy functions with many local linearizations around stochastic perfect-foresight paths, integrating over an anticipated-shock object by sigma points/quadrature and optionally over regimes via a discrete localization index. With Gaussian shocks and linear measurement equations, this yields a Gaussian-mixture state-space model amenable to Rao–Blackwellized particle filtering. We provided derivations, convergence

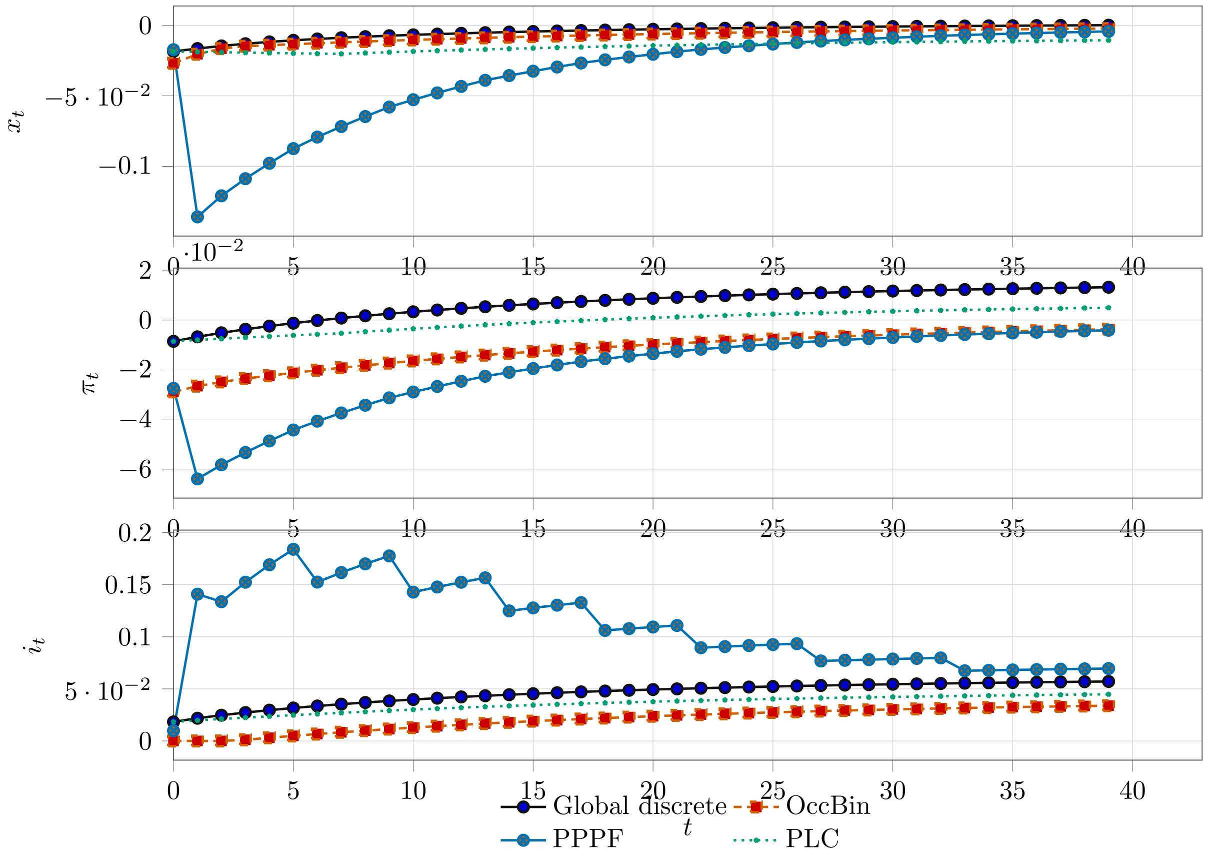

Figure 2: NK–ELB impulse responses (reference implementation): discretized global benchmark versus OccBin and PPPF-mixture approximations.

statements for quadrature and localization refinement, and stability-based links from kernel error to likelihood error. A New Keynesian ELB benchmark offers a natural comparison to the OccBin and PLC/COPF literatures.

A Deriving the conditional linear system via the implicit function theorem

This appendix provides a derivation of (3) from the stacked residual system. Let Y denote the stacked unknowns over the horizon excluding the pinned initial state:

$$Y = (x_{t+1}, x_{t+2}, \ldots, x_{t+H-1}) \in \mathbb{R}^{(H-1)n_x}.$$

Write the stacked equilibrium conditions as

$$\mathcal{F}(Y; x_t, \omega_t, \theta) = 0.$$

Suppose $\bar{Y}$ solves the system for given (x_t, ω_t). Assume the Jacobian with respect to Y is nonsingular:

$$D_Y \mathcal{F}(\bar{Y}; x_t, \omega_t, \theta) \text{ is invertible.}$$

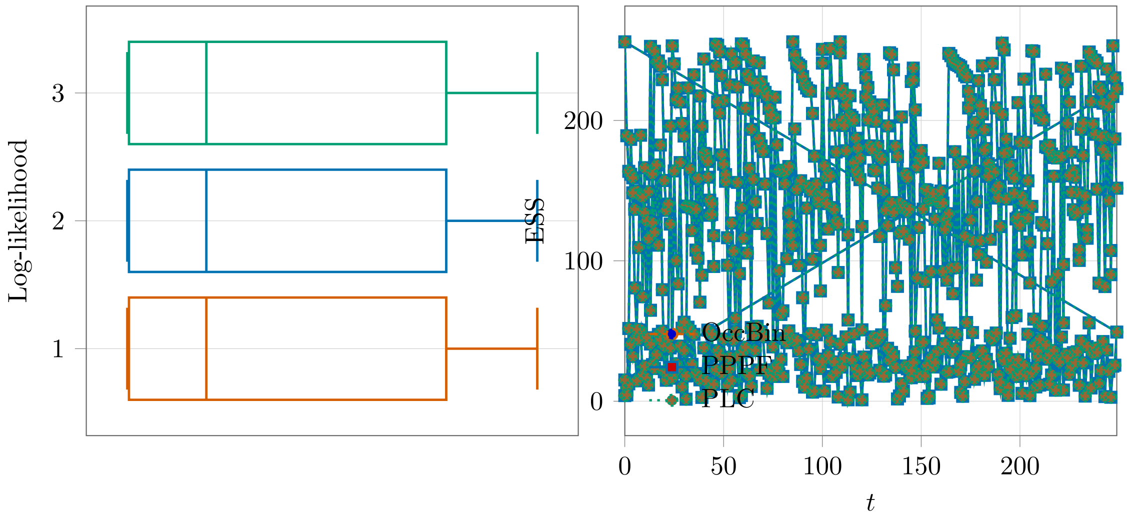

Figure 3: NK–ELB benchmark diagnostics (reference implementation): distribution of repeated log-likelihood estimates (boxplots) and ESS profile over time.

Then by the implicit function theorem there exists a local mapping $Y = G(x_t, \omega_t)$ with derivatives

$$D_x G = -\big(D_Y \mathcal{F}\big)^{-1} D_x \mathcal{F}, \qquad D_\omega G = -\big(D_Y \mathcal{F}\big)^{-1} D_\omega \mathcal{F}.$$

To incorporate an *unexpected* shock ε_{t+1}, one augments the residual function to include the contemporaneous shock in the appropriate period-$t+1$ equations, and differentiates with respect to ε_{t+1}. Because $D_Y \mathcal{F}$ is sparse block-banded (typically block tridiagonal for one-step-ahead expectations), solving for these derivatives amounts to solving a sparse linear system, which is computationally attractive in large DSGEs.

Extracting the first block row of $D_x G$ gives the mapping from x_t perturbations to x_{t+1} perturbations, yielding A_t. Similarly, the first block row of $D_\varepsilon G$ gives B_t. Plugging into $x_{t+1} = \bar{x}_{t+1} + \tilde{x}_{t+1}$ gives (3).

B Unscented-transform sigma points

For $\omega \sim \mathcal{N}(0, \Omega) \in \mathbb{R}^d$, the basic sigma-point set uses $2d+1$ points:

$$\omega^{(0)} = 0, \qquad \omega^{(i)} = +\left[\sqrt{(d+\lambda)\Omega}\right]_{\cdot i}, \qquad \omega^{(i+d)} = -\left[\sqrt{(d+\lambda)\Omega}\right]_{\cdot i}, \quad i = 1, \dots, d,$$

with weights

$$w_0^m = \frac{\lambda}{d+\lambda}, \qquad w_i^m = \frac{1}{2(d+\lambda)} \ \ (i \geq 1),$$

and covariance weights $w_0^c = w_0^m + (1 - \alpha^2 + \beta)$, $w_i^c = w_i^m$. Standard choices are $\alpha \in (0, 1]$, $\beta = 2$ (Gaussian prior), $\kappa = 0$ (Julier and Uhlmann, 2004).

C Microfoundation and variants of the NK–ELB benchmark

Equations (7)–(8) arise from household Euler equations and Calvo price setting under standard assumptions; see Woodford (2003) for a textbook derivation. The ELB (10) is an occasionally binding constraint. In empirical work one often uses a shadow-rate specification or introduces additional wedges, but the essential computational feature is the kink introduced by the max operator.

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